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„Проф. Любомир Андрейчин“ при Българската академия на науките
се организира по случай 80-годишнината от основаването на Института.**

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EXISTENTIALS, MODALS AND THE ONTOLOGY OF STATES

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Abstract. I discuss holistic and partial predicate ontologies, with focus on the realization of static spatiotemporal predicates in Russian and Bulgarian. I follow the distribution of three constructions with dative case marking on the semantic subject. Modern Bulgarian and Modern Russian share only one of them – dative-predicative structures (DPS), while Russian and Old Slavic languages also use dative-infinitive structures (DIS) as the main cue of expressing the meaning of external modality. The third type – hybrid predicates headed by a negative existential – is a bridge construction between DIS and DPS. Hybrid predicates like *негде спать* (no place to sleep), *некого винить* (no one to blame) originated in Old Russian as embedded DIS sentences, but drifted towards DPS. They are partly integrated in the ontology of the Modern Russian DPS construction and behave as modal predicatives with a number of deviating features.

Key words: *natural language ontology, Davidsonian states, modals, existentials, predicatives, Russian language, Bulgarian language, biclausality*

1. Natural language ontology

1.1. Davidsonian states, spatiotemporality and holistic ontologies

Natural language ontologies distinguish between static and dynamic situations. Dynamic situations consist of non-homogeneous phases and involve a transition from p to $\sim p$. Vendler (1957) lists three types of dynamic situations – a) activities, cf. *run, drive*, b) accomplishments, i.e. incremental or gradual predicates, cf. *build a house*, c) achievements, i.e. predicates of an instantaneous transition, cf. *notice*. These three are opposed to a single class of states. Vendler's classification does not capture the distinction between two types of static situations, from the 1960's on called Kimian and Davidsonian states (Maienborn 2007). Pre-theoretically, Kimian states can be conceived as atemporal things, results of measurement, assessments or evaluations, cf. *These tomatoes weigh 3 kilos*, while Davidsonian states are spatiotemporal things, which hold during a period of time, cf. *These tomatoes were already rotten, when John saw them*. If p is a Davidsonian state and holds in some locus during an interval starting from t_0 and ending in t_n , p is true in this locus for every time point $t_i \in \{t_0 \dots t_n\}$, so that p consists of homogeneous phases (Davidson 1980). Dynamic predicates do not fit the homogeneity criterion, while Kimian states and one-place nominal/adjectival predicates like *These trees are big, John is intelligent, John is a teacher* do not fit the spatiotemporality criterion. Trees or tomatoes as physical objects exist during a limited time; John is mortal and will no longer be a teacher after the end of his service, etc. However, the distinction between spatiotemporal and non-spatiotemporal things is crucial in the perspective of natural language ontology, cf. (Коева / Koeva 2021). It has been implemented in general predicate taxonomies like (Булыгина / Bulygina 1982; Селиверстова / Seliverstova 1982), which combine the Davidsonian notion of spatiotemporality with the analysis of dynamic predicates in the spirit of Vendler (1957).

1.2. The SLP vs. ILP distinction and partial language ontologies

Davidsonian taxonomies leave a possibility of classifying states into different types. This is done in (Булыгина / Bulygina 1982: 82 – 85) and (Селиверстова / Seliverstova 1982: 93 – 97). These authors distinguish spatiotemporal vs. non-spatiotemporal static situations: the latter, called 'свойства' ('svoystva' – properties) or 'качества' ('kachestva' – attributes) are analyzed as names of properties abstracted from any referential situations. The Russian terminology goes back to Ščerba (Щерба / Ščerba 1928), who

suggested that spatiotemporal static predicates dubbed ‘состояния’ (‘sostoyaniya’ – states) are expressed in Russian by a special word class called ‘category-of-state’¹. The same distinction was reintroduced by Carlson (1977) under the cover terms ‘stage-level-predicates’ (SLP) vs. ‘individual-level predicates’ (ILP), cf. Kratzer (1995). Carlson’s SLP class is arguably identical with Davidsonian and Ščerbian states (Zimmerling 2021c). The main difference is however that Carlsonian / Ščerbian approach only brings about *partial language ontologies* and aims at finding diagnostic contexts for SLP predicates in specific world’s languages, cf. En. *There are firemen available* (SLP) vs. **There are firemen altruistic* (ILP), or word classes associated with SLP semantics. It can also be used for interpreting selected grammatical constructions or syntactic features ($\pm$ agreement, direct/oblique case-marking, etc.) as cues of expressing SLP.

2. Dative-predicative structures and modals

2.1. Dative-predicative structures in Russian and Bulgarian

The SLP meaning can be expressed both by verbs and non-verbal predicates across the world’s languages. In Modern Russian and Modern Bulgarian, most SLP predicates have non-verbal morphology (Стаменов / Stamenov 2021) and are linked with copular sentence patterns, notably with dative-predicative structures (DPS). DPS sentences follow the pattern $N_{\text{DAT}} - V_{\text{LINK}} - \text{PRED}$. The slot PRED stands for indeclinable predicative, the copula (V_{LINK}) lacks agreement properties, and the animate semantic subject of the predicative is marked with the dative case ($^{+ANIM}N_{\text{DAT}}$). DPS predicatives, i.e. elements licensing DPS structures, constitute a lexical class both in Russian and Bulgarian. The substantial proof comes from the fact that the hypothesis that one can put every predicative/adverbial into the frame $N_{\text{DAT}} - V_{\text{LINK}} - \text{PRED}$ and produce a well-formed DPS sentence is false. The valency on the animate semantic subject marked with the dative case is diagnostic for all DPS predicatives in Russian and Bulgarian. The main difference is that in Bulgarian this valency is realized overtly in syntax and must be marked by a pronominal dative clitic², while Russian lacks fully grammaticalized pronominal clitics and obligatory pronominal marking of the experiencer. Moreover, all DPS predicatives in Russian can be realized without a filled N_{DAT} slot: what is crucial is that the $^{+ANIM}N_{\text{DAT}}$ slot cannot be added to the realizations of those elements that do not belong to the DPS class.

2.2. General meaning of the DPS construction in Russian and Bulgarian

It is plausible that all DPS predicatives in Russian and Bulgarian belong to a subclass of Davidsonian states called *internal states*. Internal states (SLP-INT) represent spatiotemporal situations with a priority animate experiential argument. They cannot be quantified externally: while X during some period of time can feel *bored*, another individual Y in the same locus and at the same time can be *interested*. Moreover, internal states cannot be visualized from outside: while a situation, where X feels *lonely* or *bored*, can have some external symptomatics, loneliness as a psychic phenomenon and experience of X is not equivalent to what Y can observe by looking at X³.

- (i) For each element Pr that regularly licenses DPS sentences in Russian or Bulgarian, a diagnostic context can be found, where Pr signals the diagnostic meaning of SLP-INT in a spatiotemporal situation.

While (i) is not necessarily true of all languages which have DPS sentences, preliminary findings indicate that it also holds for several other Slavic languages including Ukrainian, Slovenian and Serbian.

¹ Since Ščerba and his followers claimed that the correlation between spatiotemporality and the class of non-verbal predicatives is a unique feature of Modern Russian grammar, Ščerba’s hypothesis was ignored by semantic theory until 1980’s.

² The same condition holds for accusative predicatives like Bg. *страх / срам / грижа ме е* (to be afraid / ashamed / concerned).

³ A further prediction of this analysis is that Russian and Bulgarian DPS sentences tend to be *autoreferential* in the sense that the speaker himself is the source of information about his internal state of mind. This issue can be checked empirically by studying the ratio of DPS sentences in the first person in the actual (i.e. non-narrative) present time (Циммерлинг / Zimmerling 2021b).

2.3. The ontology of the Russian DPS construction

Since Russian and Bulgarian DPS constructions are selective and block elements incompatible with the taxonomic SLP-INT semantics, one can build ontologies for these constructions and explore their coverage in corpus data. This task was first solved for Russian in (Циммерлинг / Zimmerling 2017), where an ontology consisting of 15 classes was proposed. Each class is linked to a thematic group of DPS predicatives: CLASS 1 – physical states; CLASS 2 – modals; CLASS 3 – emotional states; CLASS 4 – moral evaluations; CLASS 5 – comfort; CLASS 6 – aptitude/inaptitude; CLASS 7 – internal need; CLASS 8 – compliance to the task; CLASS 9 – performance ease; CLASS 10 – willingness to perform; CLASS 11 – general assessment; CLASS 12 – (ir)relevance; CLASS 13 – efficiency; CLASS 14 – sensory and intellectual reactions; CLASS 15 – parametric features. A detailed ontology is required because of the marked tendency of DPS predicatives to form quasi-synonymic lists within the classes specified above.

A test vocabulary of 422 stimuli was double-checked both on a group of native speakers, who valued their readiness to use the test stimuli in a DPS context, and on the Russian National Corpus. This experiment aimed at finding the core of the active DPS vocabulary and establishing the language-internal variation across the speakers of Russian. The Russian test vocabulary was divided into a main list (342 stimuli) and a reserve list (80 stimuli): the latter included some fillers and elements which were initially considered less likely to be licensed in a DPS context. The speakers accepted a minor part of the stimuli from the reserve list and rejected some stimuli from the main list (mostly as old-fashioned). Otherwise the subdivision of the DPS vocabulary into the main and additional lists proved to be instrumental. The experiment helped to establish the *core* of the DPS construction, i.e. its ‘shared lexicon’ and its *extension*, i.e. the variable part produced by the speakers based on similar principles of semantic selection. All 15 classes include both the shared lexicon and the variable part. The speakers of Russian normally can express generalized SLP-INT meanings like ‘X is not well’, ‘X is ashamed’, ‘X does not care p or ~ p’ not by a single lexeme but with a quasi-synonymic set of DPS predicatives.

The ontology of the Russian DPS construction does not cover situations implying symptomatic activity of the animate dative argument and semantic control on its part. Therefore, sentences like **мне гневно* (I am angry) or **мне взволновано* (I am excited) are rated by the speakers as unacceptable.

2.4. The ontology of the Bulgarian DPS construction

In (Ivanova, Zimmerling 2019), the same double-check procedure was applied to the Bulgarian DPS construction. The analysis has shown that in spite of the fact that the vocabulary of the Russian DPS construction is larger, the Bulgarian DPS construction has a broader ontology. All 15 classes of the Russian DPS ontology can be projected to Bulgarian, while Bulgarian has three additional classes of DPS predicatives lacking exact parallels in Russian and a fourth class containing accusative predicatives. The first 15 classes, i.e. the shared part of DPS ontology, are represented by 247 test stimuli: no subdivision into main vs. reserve lists was undertaken for Bulgarian classes 1 – 15. These classes have the following coverage. CLASS 1 – physical states (28 items): *задушно / хладничко / сънливо ми е* (it feels stuffy / cool / sleepy to me); *жега ми е* (it feels hot to me). CLASS 2 – modals (16): *невъзможно / немислимо ми е* (it is impossible / unthinkable for me); *време ми е* (it is time for me); CLASS 3 – emotional states (67): *обидно / неспокойно / тежко ми е* (I feel insulted / restless / troubled); *жал ми е* (I feel sorry); CLASS 4 – moral evaluations (9): *гузно / неловко ми е* (I feel shamefaced / uneasy); *хак ми е* (I feel I deserve something negative); CLASS 5 – comfort (6): *удобно / комфортно ми е* (I feel cosy / comfortable); *по път ми е* (it is on my way); CLASS 6 – aptitude/inaptitude (2): *неуместно / нелепо ми е* (it is inappropriate / ridiculous for me); CLASS 7 – internal need: (2) *спешно / потребно ми е* (it is urgent / necessary for me); CLASS 8 – compliance to the task (9): *привично / присъщо / скъпо ми е* (it is customary / usual / expensive for me); *по джоба ми е* (it is affordable for me); CLASS 9 – performance ease (7): *тежко / лесно / непосилно ми е* (it is hard / easy / unattainable for me); CLASS 10 – willingness to perform (5): *по сърце ми е* (it is close to my heart); *не ми е до X* (I don’t feel like /dealing with/ X); CLASS 11 – general assessment (27): *нормално / непоносимо ми е* (it is normal / unbearable for me); *супер / кеф / ок / гот ми е* (I feel superb / delighted / OK / cool); CLASS 12 – (ir)relevance (6): *важно / безразлично ми е* (it is important / indifferent to me); *все едно ми е* (I couldn’t care less); CLASS 13 – efficiency (4): *изгодно / вредно / здравословно ми е* (it is beneficial / harmful / healthy for me); CLASS 14 – sensory and intellectual reactions (20): *съмнително / глупаво / безинтересно / странно ми е* (it is doubtful / silly / unin-

teresting / strange to me); CLASS 15 – parametric feature (39): *шумно / просторно / късно / рано ми е* (it is loud / wide / late / early for me). Some of these classes are small and lack high frequent items. Still, they cover the same domains as their Russian counterparts.

The additional three classes total 45 items and cover the situations where Russian speakers normally do not use DPS sentences. CLASS 16 – states of mind indicating a symptomatic activity of a person (9 items): *ядно ми е* (X is in the evil mood), *днес ми е много бърливо* (I am in the mood of chatting a lot today); CLASS 17 – lexicalized participles (8 items): *потискащо ми е* (it is overwhelming for me), *смачкано ми е след нашия разговор* (I feel depressed after our conversation); CLASS 18 – metaphoric transpositions of the names of properties and external states into the sphere of the animate subject (28 items): *дъждовно / облачно / оптимистично ми е* (I feel rainy / cloudy / optimistic). Finally, CLASS 19 includes accusative predicatives denoting strong affections (12 items): *страх / срам / яд / гнус ме е* (I feel fear / shame / anger / disgust).

Bulgarian licenses SLP-INT predicatives in situations involving symptomatic activity of a person and/or some degree of semantic control on his/her part, see especially the additional classes 16 and 17 and accusative predicatives from CLASS 19. It also marginally licenses the transition from external states (SLP-EXT) to internal states (SLP-INT), see CLASS 18. In this respect, the Russian DPS construction is more restrictive according to the corpus study (Циммерлинг / Zimmerling 2021b).

2.5. Modal DPS predicatives

Modal predicatives (CLASS 2) belong to the shared part of the DPS ontology. This class is more prominent in Russian. It includes two elements with the highest frequency from all Russian DPS predicatives – *надо* (necessary) and *нужно* (needed) and totals 44 items (12.85%) from the main list of stimuli used in the experiment (Циммерлинг / Zimmerling 2017). Four CLASS 2 items (*невместно, непригоже, повадно, некорректно* – inappropriate, unsuitable, usual, incorrect) no longer represent the actual usage: the native speakers rejected them. At the same time, part of the speakers accepted one stimulus (<мне> *обязательно* – <I> must) from the reserve list: with this correction, the ratio of the CLASS 2 items in the active DPS vocabulary is 11.98% (41 / 342).

In the Bulgarian list of DPS stimuli, CLASS 2 items make up 6.47% (16 / 247). The native speakers rated low six of them – *не за описване* (not to be dismissed), *недостойно* (dishonourable), *достойно* (honourable), *не за вярване* (unbelievable), *немислимо* (unthinkable), *неприлично* (indecent) – in a combination with a dative experiencer, despite the corresponding examples are registered in some lexicographical descriptions. This gives the corrected ratio 4.04% (10 / 247).

3. Hybrid modal existentials and related constructions

3.1. Russian Neg-words as DPS predicatives

Russian has several idioms with a modal-existential meaning. They are used in syntactic contexts resembling DPS, cf. *мне **нигде** спать* (I have **nowhere** to sleep), *мне **некого** винить* (I have **no one** to blame), *мне **ничего** делать* (I have **nothing** to do), *мне **никогда** читать книги* (I have **no time** to read books) (Fortuin 2014). The highlighted *neg*-words consist of a general negation *не* ‘not’ merged with a *wh*-word, cf. *где* (where), *кого* (whom_{ACC}), *чего* (what_{GEN/ACC}), *когда* (when), but have a far more complex argument structure with a silent existential operator, as shown by Apresyan and Iomdin (2010). The dative subject argument is almost always animate, as with DPS predicatives with different morphology. All Russian *neg*-words can be realized overtly without a filled N_{DAT} slot, but the same condition holds for all DPS predicatives. The finite position V_{LINK} is either taken by copular forms of *быть* (be), including the zero copula in the present tense indicative, or by other copulas like *стать*, cf. *Мне **стало** **нигде** спать / **никогда** читать книги* (I have nowhere to sleep / no time to read books) (Апресян / Apresyan 1996: 530). These facts suggest that from the viewpoint of surface syntax, Russian predicative *neg*-words behave as DPS predicatives. There are reasons to put them in CLASS 2, which hosts modal predicatives (Циммерлинг / Zimmerling 2021a: 278). Like other CLASS 2 items, *neg*-words denote static spatiotemporal situations (SLP). If the N_{DAT} slot is filled by an overt subject argument, the sentence expresses the diagnostic meaning of internal state (SLP-INT).

The modal meaning signaled by Russian *neg*-words is a composition of the alethic (in a different terminology – external) operator ‘possible’ ($\Diamond$) with the negation ($\sim$): ‘ p is not possible’ or ‘it is not possible that p ($\sim \Diamond p$). The proposition p is expressed by the infinitival complement. E.g. ‘It is not possible that X has a place to sleep’, ‘It is not possible that X has any time to read books’.

The negation ($\sim$) has a wider scope than the alethic operator ($\Diamond$), which has a scope over the proposition / event p . However, there is an extra shell between p and $\Diamond$: what is negated, is the *existence* of an event e , which has sub-event p expressed by a propositional complement. The existential meaning of Russian *neg*-words is confirmed by their non-negative (affirmative) correlates, which use the overt existential marker **есть**: *мне есть, где спать* (I **have** a place to sleep), *мне некого винить* (I have **no one** to blame), *мне нечего делать* (I have **nothing** to do), *мне некогда читать книги* (I have **no time** to read books). If a sentence like *(мне) негде спать* is a natural language negation of the sentence *(мне) есть, где спать*, the positive variant has the structure $\exists e P(e)$, while the negative variant has the structure $\sim \exists e P(e)$ or $\sim \Diamond \exists e P(e)$: it is impossible ($\sim \Diamond$) that there is such a place (e), where X can sleep ($\exists e P(e)$). A modal-existential interpretation of the sentence *некого винить в кризисе* is ‘it is **impossible** that any individual X existed, whom **one could blame** for the crisis’, while a purely existential interpretation of this sentence is just ‘**there is no such X** who is responsible for the crisis’. I will henceforth refer to sentence like *негде спать, некого винить* as hybrid modal existentials.

From the viewpoint of logic, either the modal or the existential component of the hybrid Russian construction can be eliminated as redundant. However, in the perspective of natural language ontology, a reductionist approach to the hybrid construction is not optimal. The in-built modal component is essential for the surface syntax of Russian *neg*-words as DPS predicatives: they show the diagnostic SLP-INT semantics, combine with copulas and make up morphologically inseparable units. The in-built existential component is essential for the assessment of their deviating features. The negative and non-negative hybrids have the same logical form, but different morphosyntax: while predicative *neg*-words only combine with copulas, non-negative hybrids require a non-copular form of the verb ‘be’.

3.2. Bi-clausality and Russian *neg*-words

The original version of Apresyan & Iomdin’s approach explores the formalism of dependency grammars, but an equivalent result can be achieved by using constituency grammars or generalized phrase structure grammars. The notions of clause, bi-clausality and raising are essential for the analysis.

- (ii) [_{ExistP} EXIST.NEG [_{TP} AUX] [_{InfP} Inf Wh]]

Hybrid modal existentials project a bi-clausal argument structure, with general negation and the existential operator in the main clause and the *wh*-word in the embedded clause. The *wh*-word raises to the main clause, merges with the negation and forms a morphological unit. The bi-clausal analysis explains the ambivalent relation of the infinitive and the *wh*-word. At the first glance, hybrid *neg*-words like *негде* (nowhere), *некого* (no one) incorporating *wh*-words like *где* (where), *кого* (whom) govern their infinitival complements: *негде_{ADV} спать_{INF}* (nowhere to sleep), *некого_{ACC} винить_{INF}* (no one to blame). However, if one considers the syntactic relation of the infinitive to the *wh*-word, the latter must be recognized as a complement of the infinitive, not its head, since pronominal *wh*-words like *кого_{ACC}* get the case from the infinitive: *винить_{INF} кого_{ACC}*. This puzzle is solved if one assumes that *wh*-words occur in the syntactic structure of hybrid existentials twice, but the lower copy in the embedded infinitival clause is deleted after the *wh*-word raises to the main clause and merges with the general negation *не*. The superscript *не*² in the notation of examples (1) – (2) below emphasizes that the general negation *не* is a hidden syntactic head or a hybrid existential operator in terms of Apresyan and Iomdin (Апресян, Иомдин / Apresyan, Iomdin 2010).

1. Ване_{DAT} **не²кому**_{PRED} [_{InfP} читать_{INF} нотации_{ACC} **кому**_{DAT}].
‘There are no people to who(m) John can read lessons.’
2. Ване_{DAT} **не²кого**_{PRED} [_{InfP} винить_{INF} в кризисе **кого**_{ACC}].
‘There are no people who(m) John can blame for the crisis.’

In (1), the lower copy of *кому* ‘to whom’ is a dative object of the ditransitive verb *читать кому-л. нотации* ‘to read someone lessons’. In (2), the lower copy of *кого* ‘whom’ is an accusative object of the verb

винить кого-л. в кризисе ‘to blame someone for the crisis.’ Note that all Russian predicatives lack case and agreement morphology (Zimmerling 2021c). It is therefore predictable that an argument *wh*-word like *кого* (whom), *кому* (to whom) loses morphological case if it raises to a non-argument position and makes up a hybrid predicative like **не²кому**_{PRED}, **не²кого**_{PRED}. If however an argument *wh*-word raises to an argument position, it preserves its case features. This raising option is available only for dative pronouns, cf. (3).

3. Ване_{DAT} **не²кому**_{DAT} [_{InfP} читать_{Inf} нотации_{ACC} **кому**_{DAT}].
 ‘There are **no people** who can read lessons to John.’

Unlike (1) – (2), sentences like (3) are incompatible with a DPS interpretation. They instantiate a different pattern – dative-infinitive structures (DIS), where the infinitive is the lexical head of a non-finite clause.

3.3. Bulgarian equivalents of Russian hybrid existentials

Bulgarian lacks hybrid existentials like *некому*, *некого*, *негде*. The *wh*-words do not merge with the general negation. Non-negative and negative existentials headed by *има* (there is) and *няма* (there is not) are built in the same way and include an embedded finite *да*-clause. Penchev (1984, 1993, 1998) however proposed that Bulgarian *wh*-words can raise out of *да*-clauses and be fronted before the complementizer *да*.

4. **Има**_{EXIST} / **няма**_{NEG.EXIST} *къде* [*да спя къде*].
 ‘There is / there isn’t a place where one can sleep.’

If this analysis is correct, Bulgarian has *wh*-raising. An alternative solution is to treat the complementizer *да* as an enclitic or enclitic-like element: in this case, the *wh*-word always remains in the embedded clause, cf. the notation in (5).

5. **Има**_{EXIST} / **няма**_{NEG.EXIST} [**да къде да спя*].

Whatever analysis is chosen, the Bulgarian syntax of existential predicates is more transparent than the Russian, and the bi-clausal structure is realized overtly both with affirmative and negative existentials.

3.4. Pure modal predicatives with neg-words

One Russian *neg*-word, *не¹чего₂* (one does not have to, one should not) has lost the existential meaning and acquired features of a standard modal DPS predicative. This item consists of the negative prefix *не¹*- and the frozen genitive/accusative form of the inanimate *wh*-word *что* (what). *Не¹чего₂* must be separated from its homonym, the hybrid predicative *не²чего₁* (there is no such X, one can operate with), where *не²* is a negative existential operator. The diagnostics comes from verbal dependencies. All realizations of the hybrid *не²чего₁* are bi-clausal and include a *wh*-word in the embedded infinitival clause, while its case form is predicted by the valency frame of the verb. Cf. *хвалить кого-л.* (to praise someone) in (6a) and its Bulgarian translation (6b).

6. a. [_{TP} **Не²чего₁**_{PRED} [_{InfP} хвалить *чего*_{ACC}]].
 ‘There is nothing to praise.’
 b. **Не съществува такъв X**, който да може да бъде похвален.

Contrariwise, the purely modal *Не¹чего₂* cannot be decomposed to elements base-generated in clauses of a different level. It is impossible to reconstruct the source of the *wh*-word *чего* in (7a), since the verb *хвалить* ‘praise’ already has its object slot filled: the *wh*-word *чего* and the NP *студентов* cannot fill in one and the same position simultaneously. The Bulgarian translation (7b) explicates the modal component *не бива да*, not the existential component *не съществува* as in (7a).

7. a. **Не¹чего₂**_{PRED} [_{InfP} хвалить *студентов*_{ACC}].
 ‘One **should not** praise the **students**.’
 b. **Не бива** да се хвали X.

Hybrid and pure modal *neg*-words differ regarding the possibility of reconstructing their non-negative correlates. All hybrids like *не²чего₁* in (6) have affirmative correlates with *есть*, while the pure modal predicative *не¹чего₂* lacks a correlate with *есть*. The sentence (8a) is ill-formed and can only be repaired by replacing the case form *чего* by *за что*, cf. (8b).

8. а. **Есть чего хвалить студентов.*
 ‘There are things the students can be praised for.’
 б. *Есть за что хвалить студентов.*
 ‘There are things the students can be praised for.’

4. DPS and DIS sentences in a diachronical perspective

DPS sentences, which have the taxonomic meaning of Davidsonian states and express a broad variety of internal states (SLP-INT) in Modern Bulgarian and Russian, and DIS sentences like (3), which have the taxonomic meaning of external (alethic) modality, existed already in Proto-Slavic and are attested in Old Church Slavonic, Old Bulgarian and Old Russian texts. Bulgarian lost the infinitive and DIS sentences, while Russian is one of the few Slavic languages where the DIS construction remains productive. The DPS and DIS constructions partly overlap, since one and the same spatiotemporal situation with an animate participant can be conceptualized both as an internal state that holds during some period of time in some locus and as external pressure leaving or depriving the subject of some possibilities. Modern Russian, with few exceptions discussed by Zimmerling (2018, 2021a: 279), does not license embedded DIS clauses, which was an option in Old Slavic languages. The broad ontologies of the Bulgarian and Russian DPS construction result from a secondary process. In the Old Slavic languages, the DPS construction had a more restrictive vocabulary and a narrow ontology: according to Khodova (1980: 242 – 247), ca. one half of the entire Old Church Slavonic DPS vocabulary patterns with modals. An additional motivation for choosing the DPS construction was that the DIS construction mostly encodes external possibility or necessity, while the CLASS 2 items can express both deontic and alethic meanings.

Three examples from the 12th century “Kyeuan chronicle” show the choice the speaker of Old Russian had A.D. 1150.

9. [DIS; ± animate subject]. Изаславъ же то видивъ . ре^ѣтъ дружинѣ свои . **оуже ми_{DAT} толико доехати_{INF} с гостьми . съ Оугры и с Лахы .** а оуже дружина моѣ пострашена тогда и самъ побѣже прочъ. (Ipat., list 146 [1150])

‘Izyaslav saw it and said to his squad: “If **only I got the chance to escape with my allies, the Hungarian and the Poles**, my own squad is already scared” – and ran away.’

10. [DPS; + animate subject]. И се ему ѡвиши црѣ на ма Грецкыи вѣставетъ ратью . **и сеѣ ми_{DAT} зимы и весны нѣ лзѣ_{PRED} на конь к тобѣ всѣсти .** но вбаче ѡце твои щить и мои не розно еста . **аче ми_{DAT} самому_{DAT} нѣ лзѣ_{PRED}** а помочи коли хочеш . 1^ѣ ли тысѣчъ болша ли а то ти послю. (Ipat., list 148 [1150])

‘And <the king> declared him: “The Greek emperor attacks me now, **and I cannot lead cavalry to you this winter and spring**. But since, father, your shield and my shield stand together and you want to get help, **even if I cannot <come> myself**, I will send you 10000 soldiers or more”.’

11. [Hybrid; ± animate subject]. И ре^ѣ льсть есть . **быти_{INF} ту Изаславу_{DAT} нѣ²гдѣ_{PRED} полкы свои в горахъ .** и за тѣмъ весь полкъ . Изаславъ ицѣли (Ipat., list 146 [1150]).

‘And he said: “It’s a trick: **there is no place for Izyaslav and his troops in the mountains**”, – and therefore Izyaslav’s army got survived.’

Example (11) demonstrates a hybrid dative construction with the negative existential *нѣ²гдѣ* (no-where, there is no place). The NP *Изаславъ* could be replaced by an inanimate expression like *полкы* (troops). It is plausible that all hybrid existentials originated in Old Russian as embedded DIS structures, cf. a root DIS sentence in (9). The diachronical drift brings them closer to DPS structures, cf. the CLASS 2

predicative *нѣтъ* (it is forbidden) in (10). In Old Russian, its non-negative correlate *лѣтъ* (it is permitted) was used as well: Modern Russian lost it, and the negation *не'* in *нельзя* is no longer separable. The same asymmetry characterizes hybrid forms like *негде спать*, since their affirmative correlates are not used in the DPS construction. Thus, Russian hybrid *neg*-structures can be viewed as a bridge construction on the way from embedded DIS to modern DPS sentences.

Conclusions

We followed the distribution of three Slavic non-verbal constructions capable of expressing modal meanings. Modern Bulgarian uses only one of them – dative-predicative structures (DIS), which cover a broad variety of spatiotemporal situations with an animate subject. Modern Russian retains dative-infinitive structures (DPS) as the main cue of expressing the meaning of external modality and develops hybrid modals existentials in negative clauses. Bulgarian lost DIS sentences and did not develop hybrid existentials, but Bulgarian parallels are important for the reconstruction of the bi-clausal syntax of Russian hybrid predicates.

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